

5. As maize seeds germinate they produce the enzyme amylase which hydrolyses starch in the endosperm into maltose. This can be demonstrated in the laboratory using the following method.
- Maize seeds are soaked in water.
 - Seeds are cut in half and placed onto starch agar.
 - After 24 hours the seeds are removed and iodine solution added to the starch agar.
 - Clear zones around the position of the seeds indicate amylase activity.
 - Boiled then cooled seeds are set up as a control.
- (a) Describe how you would develop and refine this practical procedure to determine whether treatment of seeds with a 0.1 mmol dm^{-3} solution of gibberellic acid increased production of amylase. [4]

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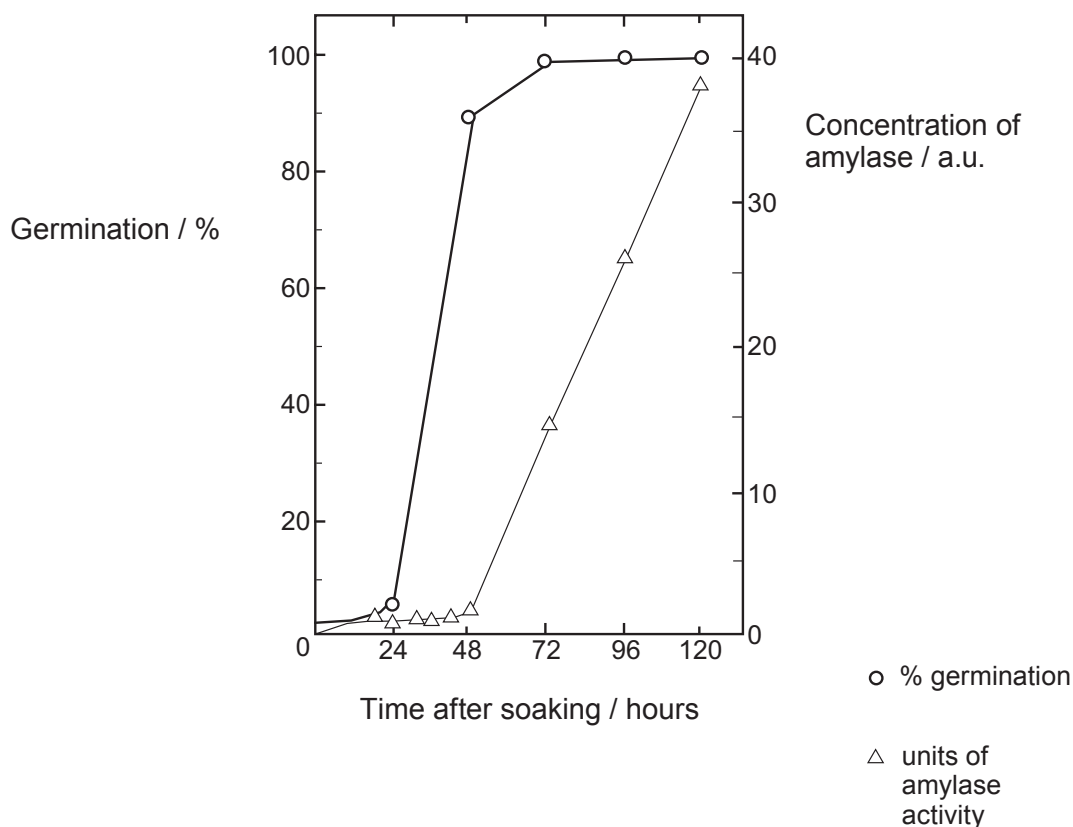
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- (b) The following experiment was carried out to test the hypothesis that gibberellic acid stimulates amylase synthesis which then causes germination.

Oat seeds were soaked in 0.1 mmol dm^{-3} gibberellic acid and placed on moist filter paper. They were kept at 23°C and the number of germinated seeds were counted at various intervals after soaking. As soon as germination occurred the seeds were homogenised (blended) and the concentration of amylase in them was determined.

The results are shown in the graph below.



Using the evidence from the graph, state whether you would accept or reject the hypothesis. Explain your answer. [2]

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